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DADIGAN: A dual attention blocks-based disentangled iterative Generative Adversarial Network for cloud and shadow removal on SAR and optical images

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ABSTRACT

The application of optical remote sensing in Earth observation missions is often hindered by cloud and shadow contamination. Leveraging the cloud-penetrating capabilities of Synthetic Aperture Radar (SAR) images, many SAR-optical image fusion approaches have been proposed and have demonstrated remarkable effectiveness in cloud removal. However, two main challenges remain in these methods: (i) Shared features, which are shared information from both modalities, and private features, which are unique to each modality, have not been fully exploited in the fusion process due to the obvious gap between multimodal images; (ii) The feature extraction processes tend to be implicit and lack interpretability. In response to these challenges and bridge this gap, we propose a novel dual attention blocks-based disentangled iterative Generative Adversarial Network (GAN) for cloud and shadow removal, named as DADIGAN. Two specialized modules are designed for the generator of DADIGAN to enhance cloud removal performance: the deep disentangled iterative network (DDIN) and the progressive dual attention fusion module (PDAFM). First, we develop an observation model based on convolutional sparse coding that explicitly models both the SAR and cloudy images as compositions of shared and private features. The Proximal Gradient Descent Algorithm (PGDA) is utilized to optimize our proposed model, implemented through successive iteration steps. These iterative steps are unfolded into the DDIN to extract the shared and private features. Second, the PDAFM employs cross-attention blocks (CABs) to fuse shared and private features using a progressive fusion strategy. Additionally, a multi-head self-attention block (MSAB) is introduced to PDAFM to capture more global information and learn contextual dependencies, thereby recovering more detailed information in regions covered by thick clouds. Comprehensive experiments have demonstrated that our proposed method delivers superior visual results and quantitative assessments under different land cover types and various levels of cloud and shadow coverage. The code for this article will be accessed at https://github.com/NUAA-RS/DADIGAN.

1. Introduction

With the continuous improvement of remote sensing detection technologies, massive remote sensing data have increasingly become the predominant method for the Earth monitoring missions [1]. Various types of remote sensing images have emerged, among which optical remote sensing images are essential for numerous applications including geographic resource surveys, ecological environment monitoring, and meteorological disaster forecasting owing to their good visual result and sufficient spectral information. However, data from the USGS and

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Landsat ETM+ indicate that clouds obscure 66% of the Earth's surface and cover approximately 35% of the land area [2–4]. The imaging mode of optical remote sensing results in inevitable contamination by cloud and shadow, diminishing the usability of these images, especially for Earth observation missions that require continuous time series of optical remote sensing images [5]. Consequently, reconstructing high-fidelity cloud-free images in optical remote sensing, which are covered by cloud and shadow, become a significant challenge.

The challenge of cloud and shadow removal in optical images is how to accurately reconstruct the spectral-spatial information of cloud-covered areas using auxiliary information. Various approaches have been developed to enhance the performance of reconstructed images, which can be broadly categorized based on the type of auxiliary information they utilize: single image restoration-based methods, multi-temporal-based approaches and multi-sensor-based approaches. Single-image restoration approaches generally assume that cloud-contaminated and cloud-less regions in the same image share similar geographical characteristics [6,7]. Based on this assumption, they use information from clear regions to reconstruct cloud-covered areas. Additionally, some single image restoration-based approaches are proposed to deal with thin cloud and haze images where surface information is not completely obscured, pixel correction techniques [8] can be applied to retain useful information while removing thin cloud and haze. However, these methods often fail to effectively deal with these cloudy images in which thick cloud-covered regions occupy a large proportion [9]. To address the issue of large areas covered by thick clouds, some researchers have proposed multi-temporal approaches [10–12], which utilize cloud-free data from other time periods in place of pixels from cloud-contaminated regions, to reconstruct the spatial–spectral information of thick clouds coverage regions. The capabilities of multi-temporal methods are often limited by the rapid geomorphologic changes over short periods of time and prolonged cloudy weather [13]. Furthermore, the computational overhead of this kind of method is generally quite large. Multisensor-based approaches aim to leverage complementary information from other satellite sensors to predict the region of cloud obscuration [14–16]. Synthetic Aperture Radar (SAR) is an active detection imaging system that captures detailed ground information by transmitting electromagnetic pulses and receiving the reflected signals [17]. This capability enables SAR to provide all-weather ground monitoring, even under persistent cloud coverage. Consequently, SAR images can offer valuable texture information of cloud-covered regions, which makes SAR-optical image fusion methods have gained popularity in the remote sensing community.

Many cloud removal methods using SAR images as auxiliary data in optical remote sensing image have been proposed over the past few years [18–25]. These methods primarily employ two fusion strategies. The first strategy [18–20,23,24], illustrated in Fig. 1(a), directly stacks the cloudy and SAR images as the input of the network to extract feature, which overlooks the unique information to each modality. The second strategy [21,22,25], shown in Fig. 1(b), extracts features from the cloudy and SAR images separately, then fuses them using some specific techniques such as concatenation or element-wise addition, which neglects the shared information between different modalities. It is worth noting that optical and SAR images have shared information because they capture ground information from the same scene. At the same time, they also have private information due to the modal heterogeneity. This can be proved by the visualization results of shared and private features from optical and SAR images under different levels of cloud and shadow coverage in Fig. 2. However, none of these approaches consider the two fusion strategies simultaneously, resulting in insufficient fusion of shared and private features. Moreover, their feature extraction process is often implicit and lacks interpretability.

To alleviate these issues mentioned above, we develop a novel dual attention blocks-based disentangled iterative Generative Adversarial Network (DADIGAN) for cloud and shadow removal. As Fig. 1(c) illustrates, a progressive fusion strategy is devised in the generator of DADIGAN which can reconstruct cloud-contaminated regions in optical images by fully exploiting the shared and private features. Overall, the generator of DADIGAN consists of three components: the deep disentangled iterative network (DDIN), the progressive dual attention fusion module (PDAFM), and the reconstruction block (RB). Specifically, we propose an observation model based on convolutional sparse coding that explicitly models both the SAR and optical images as compositions of shared and private features. The Proximal Gradient Descent Algorithm (PGDA) is used to optimize the proposed model, implemented by successive iteration steps. These iterative steps are unfolded into the DDIN to extract shared and private features. PDAFM adopts two cross-attention blocks (CABs) to fuse shared and private features based on a progressive fusion strategy. The multi-head self-attention block (MSAB) is also introduced to the PDAFM to capture global information and learn contextual dependencies to recover more detailed information under thick cloud coverage regions. Then, the RB decodes the fused features to reconstruct high-fidelity cloud-free images. Finally, the performance of the proposed method is further improved by the adversarial interaction of the GAN. Comprehensive experiments have demonstrated that our proposed method performs with superior visual results and quantitative assessments on the SEN12MS-CR dataset.

The main contributions of this paper are threefold:

- Unlike most SAR-optical image fusion approaches that rely entirely on black-box models, this paper proposes a novel dual attention blocks-based disentangled iterative generative adversarial network (DADIGAN) for cloud and shadow removal, which combines the interpretability of model-driven methods with the generalization capabilities of deep learning to effectively reconstruct regions covered by cloud and shadow.
- We show how to solve a convolutional sparse coding-based observation model to drive deep learning to extract shared and private features from SAR and cloudy images for the first time. An intuitive analysis of the feature extraction mechanism is facilitated by rationally unfolding iterative steps into neural network modules.
- A progressive dual attention fusion module is devised to adopt two cross-attention blocks to progressively fuse shared and private features from different modalities. In addition, the multi-head self-attention block is employed to capture global information and learn contextual dependencies, thereby recovering more detailed information in regions with thick cloud coverage.

For the reminder of this paper: Section 2 introduces background information about deep learning-based methods for cloud removal and deep unfolding networks. Section 3 describes our proposed network DADIGAN in detail. Section 4 presents the experiment results based on the SEN12MS dataset. Section 5 provides some discussions on the analysis. Section 6 gives the conclusion of this paper.

2. Related work

2.1. Deep learning-based methods for cloud removal

Following the vigorous improvement of deep learning [26], which has become crucial in remote sensing image processing. Over the past few years, numerous deep learning-driven approaches for optical images cloud removal have emerged, particularly those utilizing generative adversarial networks (GANs) and residual networks, which have demonstrated outstanding performance. As we all know, the generator in a GAN aims to yield a fake image that closely resembles the target image, while the discriminator strives to distinguish the generated and target images [27]. Their adversarial interaction with each other guides the GAN to reconstruct the high-fidelity cloud-free images. Bermudez et al. [28] introduced the conditional GAN-based method to reconstruct cloud-covered regions by translating SAR images into cloud-less optical images. However, its visual results were unsatisfactory on account of

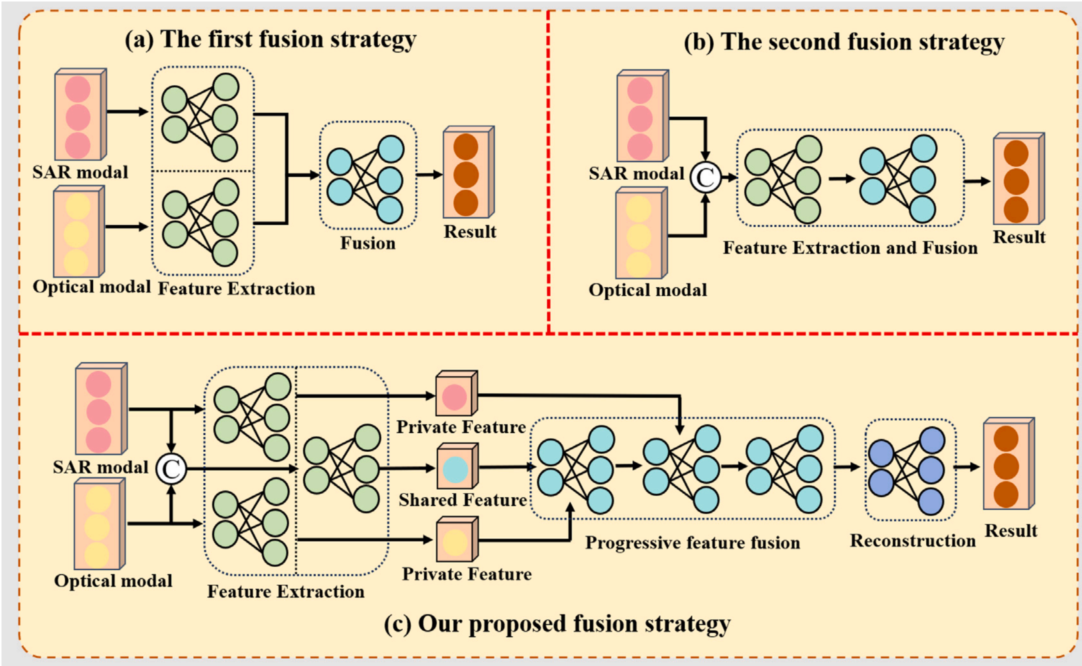


Fig. 1. The frameworks of two main fusion strategies used in the existing methods and our proposed fusion strategy.

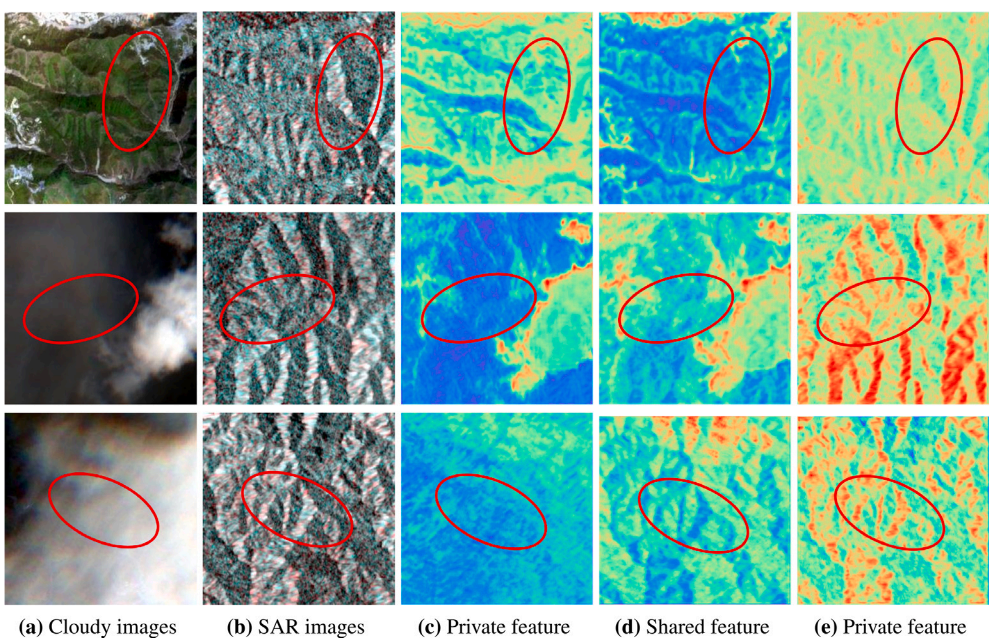


Fig. 2. The visualization results of shared and private features from cloudy and SAR images under different levels of cloud and shadow coverage, where the feature visualization results in the third and fifth columns represent the private features of the cloudy and SAR images, respectively.

the lack of prior spectral information in the SAR images. Subsequently, Grohnfeldt et al. improved the Pix2pix model [29] and proposed a new cGAN-based cloud removal method, SAR-Opt-cGAN [14], which eliminated cloud and shadow by fusing cloud-covered images with SAR images. Darbaghshahi et al. combined the aforementioned methods and developed a two-stage GAN-based network [30], that SAR images first were translated into cloud-less optical images and then cloud removal was performed. Gao et al. utilized the feature relationship between multimodal images to convert SAR images into simulated optical images. These simulated optical images, along with SAR and cloud-covered images, were used as network inputs to generate cloud-less images [31]. Moreover, they designed a local loss function to focus on the reconstruction of cloud-contaminated areas. Li et al. [21] combined the strengths of both Transformer and GAN to learn global and local relationships in remote sensing imagery for reconstructing cloud-free multispectral images (TransGAN-CFR). Furthermore, they introduced a triplet loss function to improve the performance of TransGAN-CFR. Residual networks typically use the residual block as their basic unit, which consists of several convolutional layers and a skip connection. The principle of residual blocks is to employ the skip connection to add inputs to the output, thereby addressing gradient vanishing and explosion issues occurring in deep neural networks [32]. Due to their exceptional performance, residual networks have been applied in many visual tasks, such as image restoration and super-resolution.

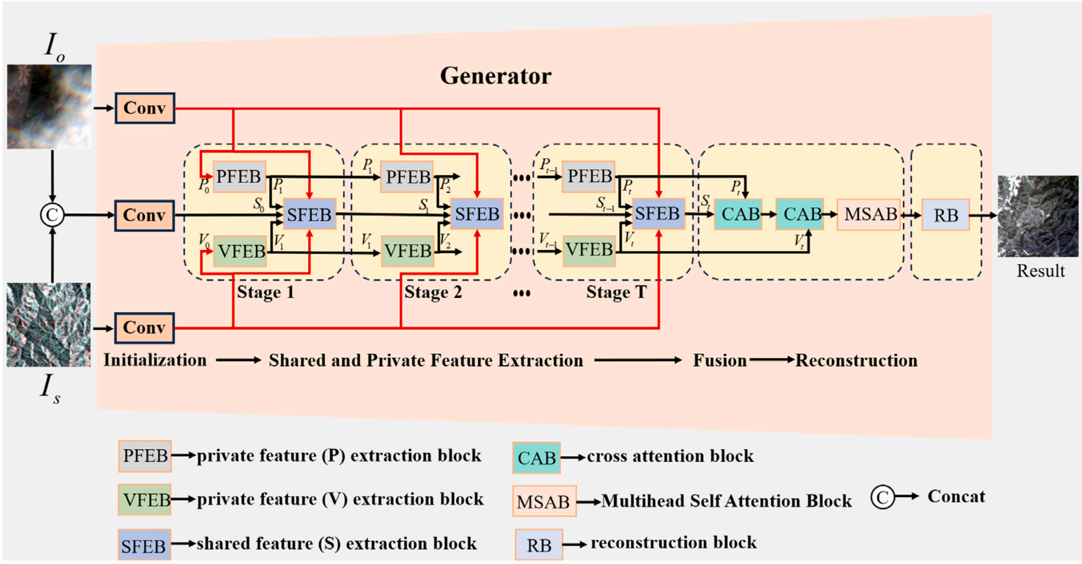


Fig. 3. The structure of the generator. The generator consists of four main components: initialization, shared and private feature extraction, feature fusion and image reconstruction. I_o denotes the cloud-covered optical remote sensing image, I_s denotes the SAR image and $Result$ represents the result of cloud removal. The size is 256 ∗ 256. The functions and abbreviations of the different modules are labeled at the bottom.

Recently, several residual network-based methods have been proposed to mitigate the challenge of cloud contamination in remote sensing images. Meraner et al. [22] leveraged residual networks to achieve superior performance in handling thin clouds removal (DSen2-CR), but the direct stacking of SAR and cloud-covered images in DSen2-CR overlooks the internal interaction correlation of multimodal images, which caused adverse reactions under thick cloud coverage. To mitigate this issue, Li et al. developed a hierarchical fusion network (HS2P) using residual blocks embedded with the channel attention mechanism. This allows HS2P to adaptively select more useful features and avoid treating channel-wise features from multimodal images equally [19].Recently, some cloud removal methods [33–35] based on transformer have emerged. These methods use the transformer to model the global correlation between the cloud-covered and cloud-free regions in optical images and the complementary correlation of different modalities, in order to reconstruct cloud-free image. Despite the decent performance of the SAR-optical data fusion methods based on these network structures, they fail to consider shared and private features from different modalities simultaneously, resulting in poor generalization.

2.2. Deep unfolding networks

Deep unfolding networks (DUNs) are implemented by unrolling a series of iterative steps derived from solving optimization problems into neural network modules, which are then trained in an end-to-end manner. Generally, DUNs combine the interpretability of model-driven approaches with the generalization capabilities of data-driven methods, significantly enhancing the interpretability of existing deep learning methods [36]. Over these years, DUNs have been extensively applied to various remote sensing image processing problems, including hyperspectral image super-resolution [37–39], hyperspectral image denoising [40], and remote sensing image fusion [41]. For the discussed image fusion problems, Xie et al. embedded the low-rank properties of hyperspectral images into a generalization model to create a new multispectral and hyperspectral image fusion model [37]. The iterative steps used to optimize the model are unfolded into an interpretable deep network, facilitating straightforward analysis of the internal mechanisms. Qu et al. designed a GAN-guided unfolded network to present the process of converting an observation model into an interpretable network through deep unfolding [41], which can achieve the better pansharpened results in an unsupervised manner. Dong et al. developed an innovative Bayesian inference model based on Gaussian noise prior for hyperspectral image super-resolution [42], enabling effective performance even when input data is polluted by the noise. It is obvious that establishing a reasonable observation model is a crucial precondition for realizing deep unfolding networks. Due to modality heterogeneity, establishing an observation model between optical and SAR images according to the above methods is challenging. Thus, we use convolutional sparse coding to explicitly model optical and SAR images as a combination of shared and private features in this paper. The reasons for constructing the observation model based on convolutional sparse coding are twofold: first, convolutional sparse coding performs well and shows great potential in a series of multimodal learning tasks [43,44]. Second, convolutional sparse coding is an extension of sparse representation [45]. Sparse representation-based optical-SAR image fusion methods for cloud removal have achieved excellent performance [15], demonstrating the possibility of cloud removal by means of convolutional sparse coding.

3. Method

3.1. Overview of the proposed network

The objectives of the proposed approach is to mitigate the inability to adequately fuse shared and private features and the lack of interpretability in the feature extraction process based on black-box models. In addition to improving the performance of the proposed method with the aid of the powerful generative capabilities of GAN, two specific designs are proposed in the generator to address the aforementioned problem. The main structure of the generator is shown in Fig. 3. More details are presented below.

We establish an observation model based on convolutional sparse coding to fully extract the shared and private features from different modalities. The observation model is expressed as follows:

$$I_o = \sum_i x_i^s * s_i + \sum_i x_i^p * p_i \tag{1}$$

$$I_s = \sum_i y_i^s * s_i + \sum_i y_i^v * v_i \tag{2}$$

where $*$ denotes convolution operation, $I_o \in R^{C_1 \times H \times W}$ denotes the cloud-covered optical remote sensing image, and $I_s \in R^{C_2 \times H \times W}$ denotes the SAR image, where H and W denote the width and height of I_o and I_s, C_1 and C_2 denote the number of channels of the cloud-covered and SAR image. x_i^s and y_i^s are the shared feature filters of I_o and I_s, and s_i denotes the shared features of I_o and I_s. x_i^p and y_i^v are the private feature filters of I_o and I_s, respectively. p_i and v_i are the private features of I_o and I_s, respectively. Assuming that all of the filter parameters are known, we can optimize the following problem to get the shared and private features.

$$
\begin{aligned}
\arg\min_{P,V,S} \frac{1}{2}\| I_o - \sum_i x_i^s * s_i - \sum_i x_i^p * p_i \|_F^2 \\
+ \frac{1}{2}\| I_s - \sum_i y_i^s * s_i - \sum_i y_i^v * v_i \|_F^2 \\
+ \lambda_s R(S) + \lambda_p R(P) + \lambda_v R(V)
\end{aligned} \tag{3}
$$

where $R(.)$ denotes the regularization term [46] related to P, V and S. P and V represent the private features from each modality, S represents the shared features from different modalities. λ_s, λ_p and λ_v denote the trade-off parameters. A series of iterative steps can be obtained by solving the model with the help of a specific optimization algorithm [47]. Then we unfold these iterative steps into some neural network modules, so as to extract the shared and private features.

Corresponding to the fusion of shared and private features, an challenge is to make a reasonable fusion strategy that can avoid the adverse reactions caused by direct stacking of features from different modalities and allow effective interactions between shared and private features from different modalities. Based on the above considerations, we design a progressive dual attention fusion module. This progressive fusion process is defined as follows:

$$
F = MSAB(CAB(CAB(P,S),V)) \tag{4}
$$

where $CAB(.)$ and $MSAB(.)$ denote the cross-attention block (CAB) and the multi-head self-attention block (MSAB), respectively. The purpose of the CAB is to fuse discrepancy information from different branches, while the objective of the MSAB is to improve the learning ability of global features to reconstruct information in regions covered by thick clouds. Finally, the fused feature F is then restored to a high-quality cloud-free image through the RB. As a result, our proposed method can completely reconstruct the spatial–spectral information of cloud and shadow coverage regions, significantly improving the interpretability of the model.

3.2. Generative network

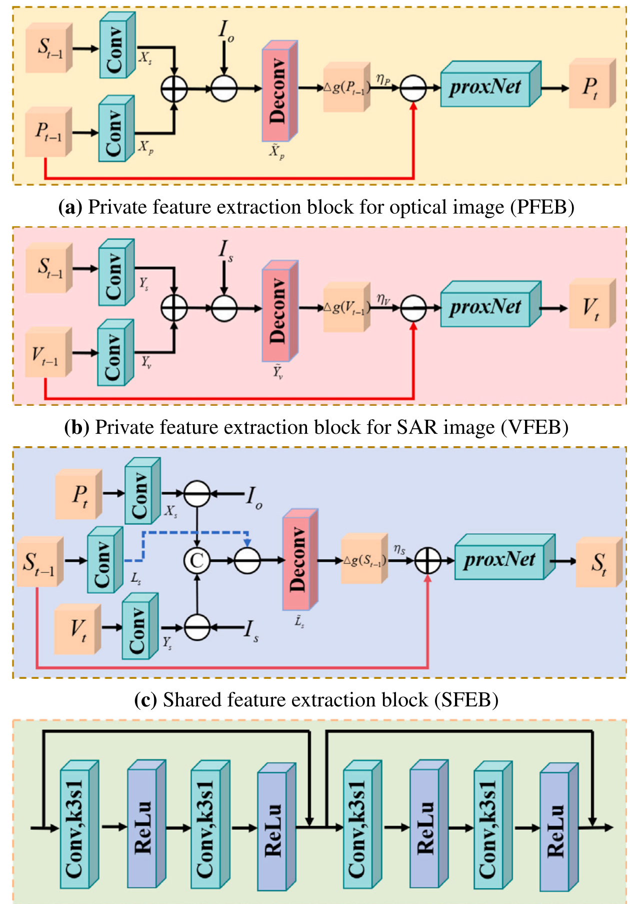
3.2.1. Shared and private features extraction module

Obviously, Eq. (3) is a multi-objective optimization problem, and it is difficult to directly solve one of the variables. The Alternating Direction Method of Multipliers (ADMM) is a core algorithm for dealing with separable block convex optimization problems in the field of mathematical optimization. It can transform multi-optimization variable problems into single-optimization variable problems through decomposition strategies. The optimization model we proposed also needs to solve three optimization variables. Therefore, inspired by the ADMM method and Deng et al. [43], we only update one variable in each step and fix the other two variables, and repeat the update alternately to solve the Eq. (3). The PGDA is adopted to address each of the three subproblems. The solution process is shown as follows:

$$
\arg\min_{P} \frac{1}{2}\| I_o - \sum_i (x_i^s * s_i + \sum_i x_i^p * p_i)\|_F^2 + \lambda_p R(P) \tag{5}
$$

$$
\arg\min_{V} \frac{1}{2}\| I_s - \sum_i (y_i^s * s_i + \sum_i y_i^v * v_i)\|_F^2 + \lambda_v R(V) \tag{6}
$$

$$
\begin{aligned}
\arg\min_{S} \frac{1}{2}\| I_o - \sum_i (x_i^s * s_i + \sum_i x_i^p * p_i)\|_F^2 + \\
\frac{1}{2}\| I_s - \sum_i (y_i^s * s_i + \sum_i y_i^v * v_i)\|_F^2 + \\
\lambda_s R(S)
\end{aligned} \tag{7}
$$



(a) Private feature extraction block for optical image (PFEB)

(b) Private feature extraction block for SAR image (VFEB)

(c) Shared feature extraction block (SFEB)

(d) The structure of $ProxNet$

Fig. 4. The network architectures of deep disentangled iterative networks (DDIN).

(1) ***Updating*** P: we solve the quadratic approximation of Eq. (4) [48] to update the private features P, which can be expressed as follows:

$$
\arg\min_{P} \frac{1}{2}\| P - (P^{t-1} - \eta_p \triangle g(P^{t-1}))\|_F^2 + \eta_p \lambda_p R(P) \tag{8}
$$

where P^{t-1} denotes the result after the $(t-1)$st iteration, η_p is the iteration step.

$$
g(P^{t-1}) = \frac{1}{2}\|(I_o - \sum_i x_i^s * s_i) - \sum_i x_i^p * p_i\|_F^2 + \lambda_p R(P) \tag{9}
$$

The optimal solution of Eq. (8) can be defined using the proximal operator as follows:

$$
P^t = prox_{\eta_p \lambda_p}(P^{t-1} - \eta_p \nabla g(P^{t-1})) \tag{10}
$$

$$
\triangle g(P^{t-1}) = X_p *^{\mathrm{T}} (X_s * S^{t-1} + X_p * P^{t-1} - I_o) \tag{11}
$$

where $*^{\mathrm{T}}$ is the transpose convolution operation which can be learned by the transpose convolutional layer, $X_p \in R^{1\times n\times n\times i}$ and $X_s \in R^{1\times n\times n\times i}$ represent the filters involved with x_i^p and x_i^s, respectively. $prox_{\eta_p \lambda_p}$ is the proximal operator that depends on the regularization term.

(2) ***Updating*** V: analogously, the quadratic approximation of Eq. (6) is used to update the private features V, which can be expressed as:

$$
\arg\min_{V} \frac{1}{2}\| V - (V^{t-1} - \eta_v \triangle g(V^{t-1}))\|_F^2 + \eta_v \lambda_v R(V) \tag{12}
$$

where V^{t-1} denotes the result after the $(t-1)$st iteration, η_v is the iteration step.

$$g(V^{t-1}) = \frac{1}{2}\|(I_s - \sum_i y_i^s * s_i) - \sum_i y_i^v * v_i\|_F^2 + \lambda_v R(V) \tag{13}$$

The optimal solution of Eq. (12) can be defined using the proximal operator as follows:

$$V^t = prox_{\eta_v \lambda_v}(V^{t-1} - \eta_v \nabla g(V^{t-1})) \tag{14}$$

$$\triangle g(V^{t-1}) = Y_v *^{\mathrm{T}} (Y_s * S^{t-1} + Y_v * V^{t-1} - I_s) \tag{15}$$

where $Y_v \in R^{1\times n\times n\times i}$ and $Y_s \in R^{1\times n\times n\times i}$ represent the filters involved with y_i^v and y_i^s, respectively. $prox_{\eta_v \lambda_v}$ is the proximal operator that depends on the regularization term.

(3) *Updating* S: we first rewrite the Eq. (7) as follows:

$$\arg\min_S \frac{1}{2}\|\widetilde{I}_o - \sum_i x_i^s * s_i\|_F^2 + \frac{1}{2}\|\widetilde{I}_s - \sum_i y_i^s * s_i\|_F^2 + \lambda_s R_s(S) \tag{16}$$

where $\widetilde{I}_s = I_s - \sum_i y_i^v * v_i$, $\widetilde{I}_o = I_o - \sum_i x_i^p * p_i$. We further simplify the first and second terms of Eq. (16), which can be expressed as:

$$\arg\min_S \frac{1}{2}\|\widetilde{I} - \sum_i l_i^s * s_i\|_F^2 + \lambda_s R_s(S) \tag{17}$$

where $\widetilde{I}$ represents the concentration of $\widetilde{I}_s$ and $\widetilde{I}_o$. Similarly, the same approach as for P and V is used to optimize this subproblem. The optimization process is shown as follows:

$$\arg\min_S \frac{1}{2}\|S - (S^{t-1} - \eta_s \triangle g(S^{t-1}))\|_F^2 + \eta_s \lambda_s R(S) \tag{18}$$

$$S^t = prox_{\eta_s \lambda_s}(S^{t-1} - \eta_s \nabla g(S^{t-1})) \tag{19}$$

$$\triangle g(S^{t-1}) = L_s *^{\mathrm{T}} (L_s * S^{t-1} - \widetilde{I}) \tag{20}$$

where $L_s \in R^{1\times n\times n\times i}$ represents the filters involved with L_s, $prox_{\eta_s \lambda_s}$ is the proximal operator that depends on the regularization term.

As shown in Fig. 4(a–c), we unfold Eqs. (10) (14) (19) into deep disentangled iterative networks(DDIN) which contains three sub-networks, *i.e.* PFEB, VFEB, and SFEB. Three sub-networks achieve the prediction of P, V, and S, respectively. The parameters of X_p, X_s, Y_s, Y_v, and L_s are learned by the convolutional layer, the size of all convolutional kernels is $3 * 3$, the number of channels is 64 (Except for L_s, the number of channels is 128). The parameters of $L_s *^{\mathrm{T}}$, $X_p *^{\mathrm{T}}$, and $Y_v *^{\mathrm{T}}$ are learned by the transpose convolution layer, which are denoted as $\widetilde{L_s}$, $\widetilde{X_p}$, and $\widetilde{Y_v}$, respectively. It should be pointed out that instead of employing a hand-crafted image prior, we use Res-Net [49] to implicitly learn the prior information $prox_{\eta\lambda}$, η is the reciprocal of the Lipschitz constant. The structure of Res-Net is shown in Fig. 4(d).

3.2.2. Progressive dual attention fusion module

Existing feature fusion methods typically use element-wise addition to fuse shared and private features from multimodal images. However, the visualization results of shared and private features in Fig. 2 show that shared features S and private features P are contaminated under thick cloud coverage. The element-wise addition operation directly transfers the features of cloud-covered regions into the fusion result, failing to accurately reconstruct the spatial–spectral information of cloud-covered areas. Therefore, as shown in Fig. 5, progressive dual attention fusion module is devised to effectively fuse shared and private features and reduce adverse reaction caused by the element-wise addition, which consists of two CABs with the same structure and different parameters and a MSAB. Specifically, the first CAB fuses the private features P of the optical image and shared features S to obtain an intermediate feature F_M. The private feature P is linearly reshaped to the query Q_p, and the shared feature S is linearly transformed to the key K_s and value V_s. A matrix multiplication of Q_p and the transpose of K_s yields the attention weight matrix, which computes the feature similarity between Q_p and K_s. This process is given as follows:

$$Attention = softmax(\frac{Q_p K_s^T}{\sqrt{d}}) \tag{21}$$

where d is the scaling factor, which avoids the problem of vanishing softmax gradient caused by the dot product result becoming too large as d increases. It should be noted that we used matrix factorization method to reduce the amount of calculation when calculating the attention weight. After that, we can remove similar features between P and S to transfer the private features P of the optical image to the intermediate feature F_M, This has the advantage of both the fusion of the discrepancy information in P and S and alleviating the feature transfer in the cloud coverage region. This process is shown below:

$$F_M = Linear(V_s(1 - Attention)) + Q_p \tag{22}$$

$$F_M = MLP(Norm(F_M)) + F_M \tag{23}$$

where $MLP(.)$ represents the Multilayer Perceptron (MLP), $Norm(.)$ denotes the linear norm operation. Subsequently, the fusion process of the intermediate features F_M and the private feature V in the second CAB is the same as the first CAB, which are defined as follows:

$$Q_v, K_f, V_f = Linear(Reshape(V, F_M, F_M)) \tag{24}$$

$$F = Linear(V_f(1 - softmax(\frac{Q_v K_f^T}{\sqrt{d}}))) + Q_v \tag{25}$$

$$F = MLP(Norm(F)) + F \tag{26}$$

The self-attention mechanism [50] captures the dependencies between pixels at arbitrary positions and pixels at other locations through a global receptive field. This capability enhances the utilization of non-local features, proving its effectiveness in image restoration tasks and showing significant improvements in reconstructing regions covered by thick clouds [51–54]. Consequently, the MSAB is integrated into the PDAFM. As illustrated in Fig. 5, the fusion result F of the CABs is fed into the MSAB to better capture the global information. The fusion result F is first put into the normalization layer to stabilize the feature distribution. Then the attention map of F is computed by the multi-head self-attention mechanism, which computes the attention result of each head separately and concentrates the attention maps of all heads. The process of the MSAB is given below:

$$Q, K, V = FW \tag{27}$$

$$[Q_1 ... Q_i], [K_1 ... K_i], [V_1 ... V_i] = Spilt(Q), Spilt(K), Spilt(V) \tag{28}$$

$$head_i = soft\max(\frac{Q_i K_i^T}{\sqrt{d}})V_i \tag{29}$$

$$Multihead = concat(head_1, head_2 ... head_i) \tag{30}$$

where W denotes the depth-wise convolution, $Spilt(.)$ represents the split function. After that, we introduce the gating mechanism $G(.)$, which is used in feed-forward neural networks (FFN) to guide the attention results. It consists of a convolutional layer with a kernel of 1*1 and a GELU activation function. Then, a Hadamard product between G(F) and the attention result M^a is performed to obtain the feature F_G. Finally, we use a skip connection to add the input F to the output to obtain the final result F_{fued}.

$$F_{fued} = MSAB(F) \odot G(F) + F \tag{31}$$

3.2.3. Reconstruction module

As shown in Fig. 6, we use residual blocks as the backbone structure of the RB, which consists of five residual blocks and a convolutional layer. Each residual block contains three convolutional layers and a skip connection, with the first two convolutional layers accompanied by a

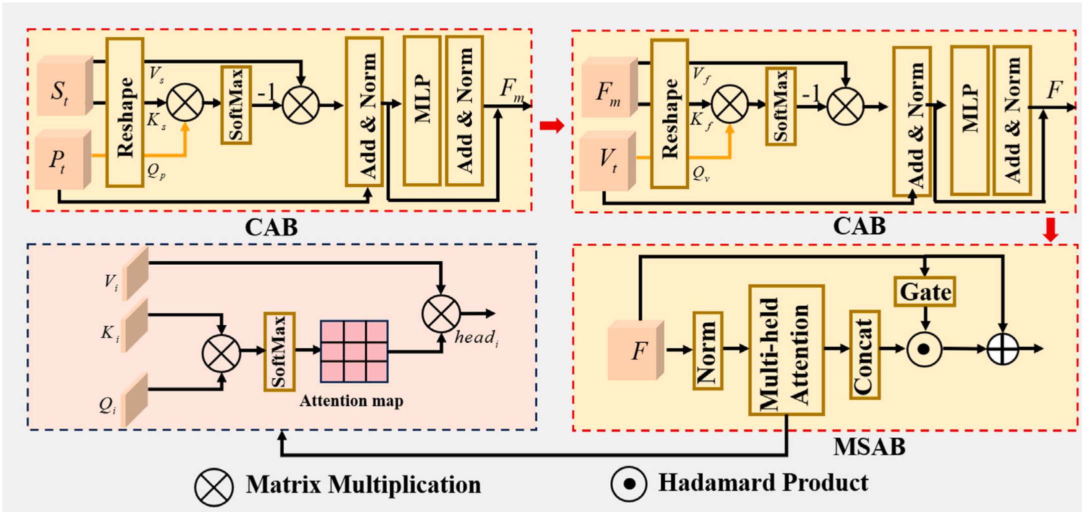


Fig. 5. The pipeline of the progressive dual attention fusion module.

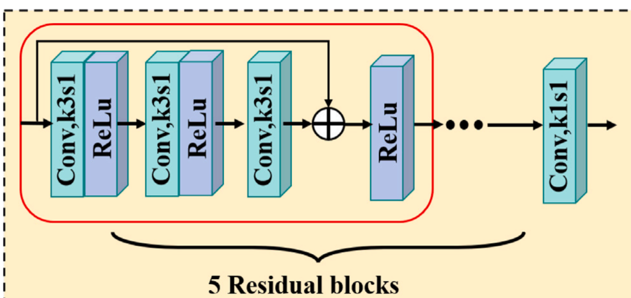


Fig. 6. The network structure of the reconstruction block.

rectified linear unit (ReLU). The size of the convolutional kernel in each residual block is 3*3, and the numbers of input and output channels is 64. The final convolutional layer has a convolutional kernel size of 3*3 with 64 input channels and 13 output channels to match the spectral dimension of the generated cloud-less image.

3.3. Discriminative network

The synthesized and real reference image are stacked as the input for the discriminator. The goal of discriminator is to determine as accurately as possible whether the synthesized image is cloud-less image or not, thereby guiding the training direction of the generator. As Fig. 7 shows, the discriminator consists of five blocks of the same structure and a convolutional layer. Specifically, each block contains a convolutional layer with the convolutional kernel of 4*4, a normalization layer, and an activation function Leaky ReLu, where the stride of the convolutional layer is 2. However, the last convolutional layer has a convolutional kernel size of 3 and a stride of 1. The output feature dimension of the ith block is $2^{4+i}, i \in [1,5]$ and the last convolutional layer outputs a 1D data to represent the result of the discriminator.

3.4. Loss function

The loss function is utilized to optimize the parameters of the proposed network and guide the training direction of the generator, which can be denoted as $\mathcal{L}_{loss}$. $\mathcal{L}_{loss}$ consists of three parts, defined by the following equation:

$$\mathcal{L}_{loss} = \arg\min_{G}\max_{D} \mathcal{L}_{cGAN}(G,D) + \lambda_1 \mathcal{L}_1(G) + \lambda_2 \mathcal{L}_{KL} \tag{32}$$

where $\mathcal{L}_{cGAN}$ denotes the adversarial loss of the conditional GAN, $\mathcal{L}_1(G)$ denotes the $L1loss$, $\mathcal{L}_{KL}$ denotes the Kullback–Leibler (KL) divergence loss [55]. λ_1 and λ_2 are the tuning parameters, which will be discussed in more detail in Section 5.3. More details about the loss function are given below. L_{cGAN} can be expressed as:

$$\begin{aligned}\mathcal{L}_{cGAN}(G,D) = {} & E_{m,n\sim Q_{data}(m,n)}[\log D(m,n)] \\ & + E_{m\sim Q_{data}(m,n)}[\log(1-D(m,G(m)))]\end{aligned} \tag{33}$$

where m denotes the cloudy image, n denotes the real cloud-free image, Q_{data} represents the data distribution, $G(m)$ denotes the reconstructed cloud-less image, and $D(.)$ represents the probability that the synthesized cloud-less image is close to the real reference image. $\mathcal{L}_1(G)$ is mainly exploited to preserve more detailed information so that the generated image is closer to the cloud-free image, the process is formulated as follows:

$$\mathcal{L}_1(G) = \frac{1}{HWC_1}\|n - G(m)\|_1 \tag{34}$$

where HWC_1 represents the number of pixels of the multispectral image. We further introduce $\mathcal{L}_{KL}$ to regularize the spectral distribution between the reconstructed and cloud-free images, which can be denoted as follows:

$$\mathcal{L}_{KL} = KL(p(G(m))\|q(n)) \tag{35}$$

where $p(.)$ and $q(.)$ denotes the probability distribution, used to enforce the spectral distribution of the generated image to be as similar as possible to that of the target image.

4. Experimental results and analysis

4.1. Dataset and evaluation metrics

To test the performance of the proposed network DADIGAN, we employ the two public datasets to train and test both our method and the comparison method.

(1) The SEN12MS-CR [56] dataset is generated by processing freely available Sentinel satellite data, supported by the Copernicus Project. All data from SEN12MS-CR is georeferenced with a ground sampling distance of 10 m and encompasses all inhabited continents and meteorological seasons, maximizing data authenticity and generalization. Specifically, the observed regions in SEN12MS-CR are primarily derived from 175 non-overlapping regions of interest (ROIs), which contain a total of 122,218 triplets. Each triplet consists of a cloudy Sentinel-2 image, a dual-polarized Sentinel-1 SAR image, and a cloud-free Sentinel-2

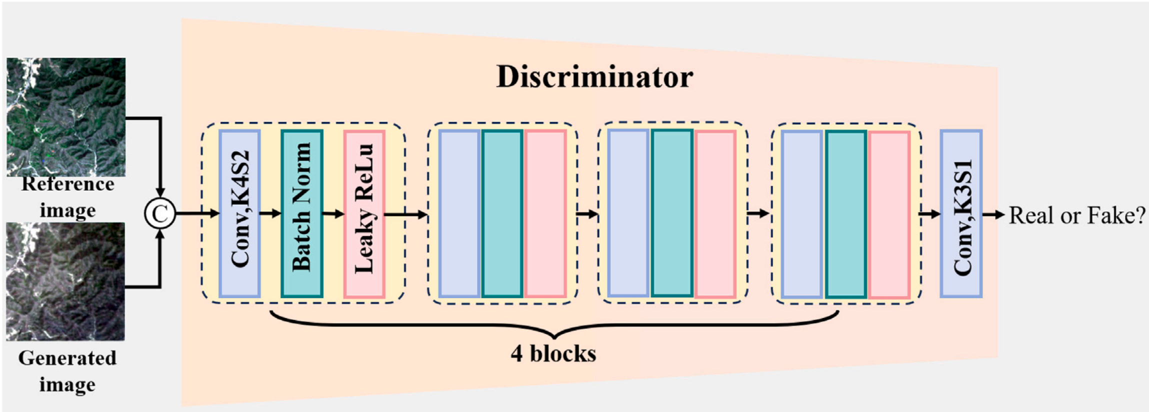


Fig. 7. The network structure of the discriminator.

Table 1
Quantitative assessment of all comparison methods.

Methods	Input data		SEN12MS-CR				SMILE-CR			
	Optical	SAR	PSNR	SSIM	SAM	MAE	PSNR	SSIM	SAM	MAE
raw	✗	✗	20.3548	0.5231	19.5748	0.0689	–	–	–	–
Pix2pix	✗	✓	26.2903	0.8024	12.2468	0.0425	29.5202	0.8112	6.8533	0.0252
McGAN	✓	✗	26.4264	0.8257	10.7564	0.0392	29.6719	0.8426	6.2799	0.0240
SAR-opt-cGAN	✓	✓	26.7542	0.8262	9.6548	0.0352	29.8542	0.8537	6.0432	0.0231
Dsen2-CR	✓	✓	26.9428	0.8324	8.9036	0.0316	31.8467	0.9246	4.9773	0.0235
GLF-CR	✓	✓	28.9427	0.8724	8.0324	0.0267	34.9343	0.9217	4.0632	0.0211
Align-CR	✓	✓	27.9523	0.8527	8.5276	0.0294	34.4367	0.9250	4.1298	0.0225
Ours	✓	✓	**30.2185**	**0.9025**	**7.1457**	**0.0231**	**35.5426**	**0.9337**	**3.5054**	**0.02017**

image, each with a size of $256 * 256$ pixels. The acquisition times of the images within each triplet are closely matched to minimize training deviation due to variations between the reference and input images. 70%, 20%, and 10% of the 175 ROIs are used for training, validation, and testing of the network, respectively. Notably, due to the different imaging principles between optical and SAR images, their values must be processed separately to meet network training requirements. The cropping ranges for the VV and VH channels in the dual-polarized SAR images are set to [−25,0] and [0,32], respectively. For optical images, values are cropped to the range of [0,10000]. Then they are all normalized to fall between 0 and 1. To fully utilize the spectral information of multispectral images and the spatial information of SAR images, we use dual polarization channel data for SAR images and 13 bands data for multispectral images.

(2) The SMILE-CR dataset [57] contains a total of 1400 globally representative image groups. Unlike the SEN12MS-CR dataset, the optical images in each image group of the SMILE–CR dataset come from Landsat −8 and MODIS, respectively. In order to satisfy the requirements of cloud removal task in this paper, we choose the SAR images from Sentinel 1, cloudy and cloud-free images from Landsat −8 form a triplet for training. Specifically, the SAR image has a spatial resolution of 10 m with 2 polarization channels (VV, VH) and the optical image has a spatial resolution of 30 m is with 6 bands. Optical and SAR images are aligned and sampled to the same size, their size is $512 * 512$. 70%, and 10% of the 1400 image groups are used for training, validation, and testing, respectively. In addition, we cropped the optical and SAR images in each triplet into four non-overlapping images of $256 * 256$ pixels to adapt the training of different networks.

Four commonly image quality metrics [58] are used in this experiment: PSNR and MAE evaluate the quality of the reconstructed image by calculating the visual error, SAM calculates the spectral similarity [59] between the reconstructed and reference image, and SSIM calculates the spectral similarity between the reconstructed and reference image. All evaluation metrics mentioned above are obtained by averaging the values across each band of the multispectral image.

4.2. Experimental setup

Our proposed method has been debugged in pytorch 12.1 and trained on dual GPUs NVIDIA GTX 4090 with the Adam optimizer [60] using the following parameters: the learning rate of the generator and discriminator is empirically set as 4×10^{-5}, the entire training process was about 200 epochs with a batchsize of 8, the tuning parameters in Eq. (32) are set as $\lambda_1 = 100$ and $\lambda_2 = 100$. Additionally, we use three convolutional layers with the kernel of 3*3 to initialize private features V_0, P_0 and shared features S_0, the iteration stage T is set to 3.

4.3. Experimental results

To validate the performance of our proposed method, we compared it with six representative methods, comprising both single-image-based methods and optical-SAR fusion-based methods. The former includes Pix2pix and McGAN, while the latter includes SAR-Opt-cGAN, DSen2-CR, CLF-CR, and Align-CR. It should be emphasized that the original codes of these comparison methods, publicly available on GitHub, are used to train the networks. Table 1 presents the quantitative evaluation results of all comparison methods and our proposed method on the two public datasets, where the block bold font represents the optimal results and underlined results indicate second best results. It is worth noting that the first row (raw) shows the test results between cloud-covered images and cloud-free images. The test set of the SMILE-CR dataset contains a certain number of cloud-free images, which may interfere with the results and are not recorded. The quantitative average evaluation results in Table 1 intuitively indicate that our proposed method achieves the best performance on the four metrics. Figs. 8 and 9 show the visual reconstruction results of all comparison methods and our proposed method on two datasets under different seasons and land cover types,respectively. The sizes of all the test image are $256 * 256$. On the whole, our proposed method effectively addresses the problem of cloud and shadow coverage under different land cover types and demonstrates good robustness across different levels of cloud coverage.

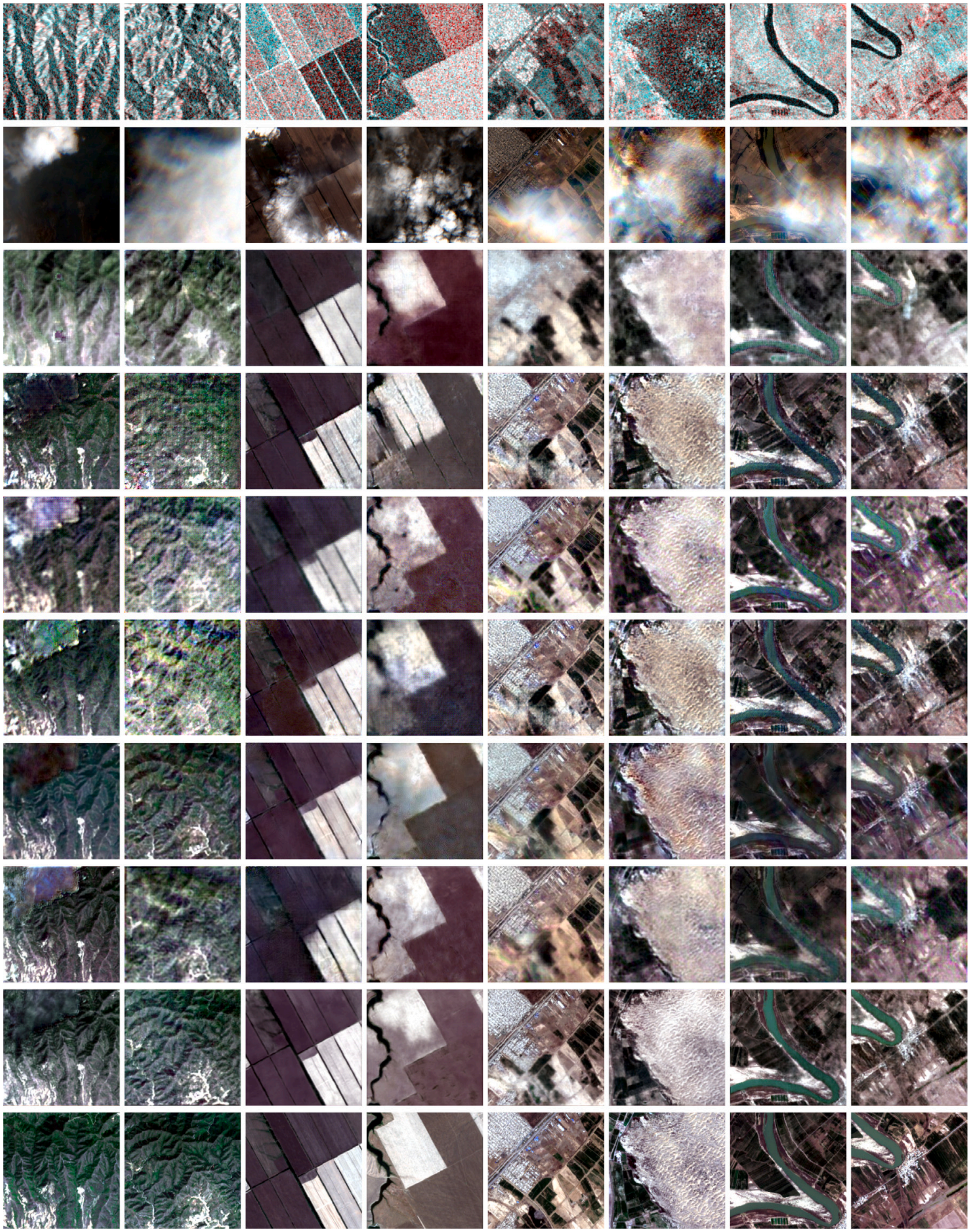


Fig. 8. Cloud and shadow removal results on the SEN12MS-CR dataset under different land cover types. For each example sample, from the top row to the bottom row represent SAR images, cloud images, the results of Pix2pix, McGAN, SAR-Opt-cGAN, Dsen2-CR, GLF-CR, Align-CR, ours, and the cloud-free images.

As shown in the results in the third row of Figs. 8 and 9, Pix2pix utilizes the prior spatial information provided by SAR images to reconstruct cloud cover areas through SAR-to-Optical image translation. However, due to modal heterogeneity, SAR images cannot fully capture the unique information presented in optical images, resulting in the disappearance of some content in the reconstruction results of Pix2pix. The relatively simple network structure of Pix2pix is incapable of learning deep features from multimodal images, which is also one of the reasons for the poor performance of the model. McGAN performs well in scenes with relatively regular distribution of geographic features, but is prone to serious blurring and noise in mountainous areas and urban areas with complex texture features. This is mainly due to the fact that the network structure is only an extension of conditional GAN and the network input lacks the spatial information under cloud-covered areas provided by SAR images, which makes it impossible to accurately learn the complex nonlinear mapping relationship between cloud-covered images and cloud-free images in areas with rapidly changing features. Under thick cloud coverage, the visual results in

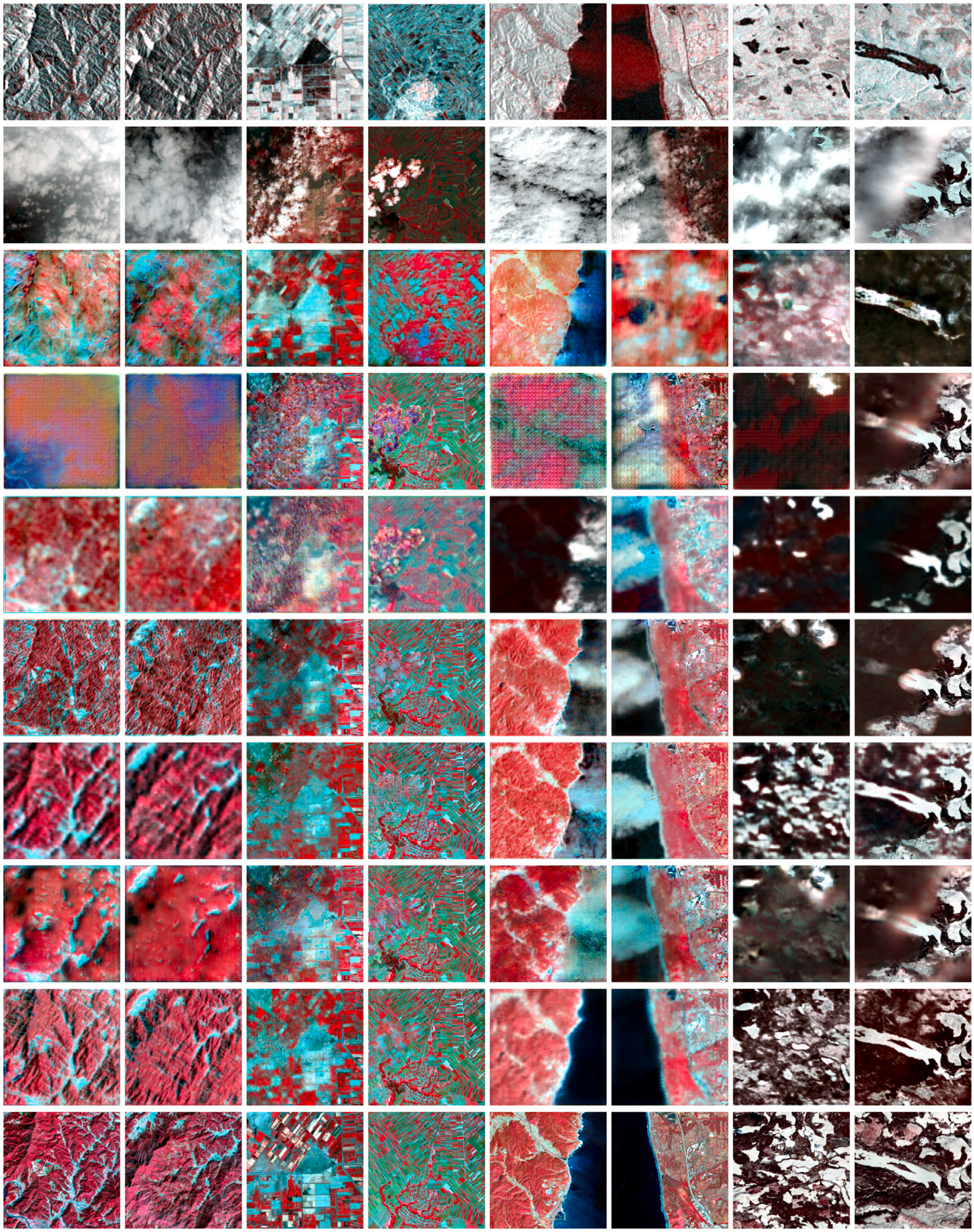


Fig. 9. Cloud and shadow removal results on the SMILE-CR dataset under different land cover types. For each example sample, from the top row to the bottom row represent SAR images, cloud images, the results of Pix2pix, McGAN, SAR-Opt-cGAN, Dsen2-CR, GLF-CR, Align-CR, ours, and the cloud-free images.

the fourth row of Fig. 9 show obvious block artifacts due to the complete lack of prior information. In contrast to Pix2pix and McGAN, which only input cloud-covered images or SAR images, SAR-Opt-cGAN tries to stack optical and SAR images as the network input, and its overall performance outperforms the results of Pix2pix and McGAN. However, SAR-Opt-cGAN is unable to adequately fuse the shared and private features of the SAR and optical images resulting in significant optical artifacts displayed in its reconstruction results of mountainous scenes, which highlights the challenge of fully leveraging the unique information provided by multi-modal images to avoid performance degradation caused by modality heterogeneity. DSen2-CR employs a residual network structure and concatenate the cloudy images into the final reconstruction results using skip connections, alleviating the challenge to a certain extent, but it still encounters spectral distortion in the visual results in the fifth row of Fig. 8 for thick cloud-covered mountainous scenes. This also indicates that stacking cloudy and SAR images as inputs fails to capture the intra-modal contextual relationships and inter-modal complementary relationships between optical and SAR images thus leading to unsatisfactory reconstruction results. CLF-CR achieves the most optimal visual results in Figs. 8 and 9 for cloud-free image reconstruction among all comparison methods, owing to its global-local fusion strategy that fully exploits the supplementary information provided by SAR images. However, an unavoidable drawback is that the transformer-based attention mechanism used by CLF-CR limits its applicability to fixed-size optical images. When the size of the cloudy image changes, the model is no longer applicable. Align-CR considers alignment errors between different resolution images but neglects the interaction between multimodal data, resulting in poor model robustness, particularly in mountainous scenes. Compared to other methods, our approach leverages shared and private features from optical and SAR images simultaneously, rather than focusing on just one. Additionally, we designed a progressive dual attention fusion module to fully integrate specific information from different modalities, enabling adaptive reconstruction of cloud-covered areas.

Besides, we further test the performance of the proposed approach under various levels of cloud and shadow coverage in the SEN12MS-CR dataset. Fig. 10 illustrates the cloud removal results from our proposed approach and several comparative methods under different degrees of cloud and shadow coverage in mountainous scenarios with complex and varied terrain, in which the cloud coverage level of the test cases in the first row progressively increases from left to right, ranging from cloud-free (first column) to complete cloud-covered (last column). The performance of McGAN gradually deteriorates as the level of clouds and shadows coverage increases. The reason for the performance degradation derives from the reduction of available cloud-free prior information in the input data. Conversely, the performance of Pix2pix is basically unaffected by changes in cloud and shadow coverage levels. Although both SAR-Opt-cGAN and DSen2-CR use optical and SAR images as inputs, the direct stacking of input data images in both methods leads to adverse reactions, resulting in performance degradation. The reconstruction results of SAR-Opt-cGAN are contaminated with noise and severe optical artifacts while Dens2-CR reconstructs only the spatial information of the cloud-covered region and suffers from severe spectral distortion. Align-CR performs relatively stable under different levels of clouds and shadows coverage, managing to reconstruct most of the spatial–spectral information in cloud-covered areas, albeit with minor spectral distortions. CLF-CR mitigates the performance degradation observed in SAR-Opt-cGAN and DSen2-CR by accounting for interactions between different modalities, achieving superior overall performance compared to other comparison methods. Our proposed method employs an improved attention mechanism to fully integrate specific information from different modalities and adopts a progressive fusion strategy to avoid adverse reactions caused by modality heterogeneity. This enables our method to produce fusion results that are closest to the reference images under various levels of cloud and shadow coverage. Even under extensive cloud and shadow coverage, the performance of our method surpasses that of CLF-CR, particularly in terms of spectral information.

5. Discussion

5.1. Ablation experiment

Each component of our proposed method plays a crucial role in reconstructing high-fidelity cloud-free images. To analyze and determine the contribution of each module to the overall performance, we performed ablation experiments on the test dataset. The detailed results and analyses are presented below.

Table 2
Quantitative assessment of different the number of stage T.

Number of stages	1	2	3	4
PSNR↑	28.5362	29.3265	**30.1205**	28.3267
SSIM↑	0.8853	0.8913	**0.9015**	0.8854
SAM↓	8.5287	8.4361	**7.1432**	7.5674
MAE↓	0.0286	0.0275	**0.0231**	0.0254

5.1.1. The impact of shared and private features

To verify the impact of shared and private features on the performance of the model, we remove shared or private features during the network training respectively, recorded as w/o PF and w/o SF. The caveat is that when only shared features are maintained, the MSAB is used and CABs are not employed in PDAFM. Similarly, when only private features are maintained, The result of their addition is sent directly to the MSAB. In general, compared to w/o PF and w/o SF, the quantitative results in Fig. 11 make it clear that the proposed approach based on the full fusion of shared and private features achieves superior results. As shown in Fig. 12, although w/o PF and w/o SF can recover most of the spatial information under thin cloud coverage, they still suffer from different degrees of spectral distortion compared with our proposed method. Furthermore, this performance gap becomes larger as more regions are covered by thick clouds.

5.1.2. The effectiveness of the PDAFM

To validate the effectiveness of the PDAFM and distinguish the contribution of the CAB and MSAB to the improvement of model performance. To be more specific, we remove the CAB or MSAB during the network training respectively, which are recorded as w/o CAB and w/o MSAB. It should be noted that the result of feature addition to feed into the MSAB directly when the CAB is removed. Compared with w/o CAB, the quantitative comparison results in Fig. 11 and the visual results under various levels of cloud coverage in Fig. 12 show that our proposed method can effectively avoid the adverse reaction caused by the element-wise addition of features and improve the performance of cloud removal. With respect to w/o MSAB, the visual comparison results in Fig. 12 show that the MSAB adopted in the PDAFM improves the reconstruction results in thick cloud-covered regions.

5.1.3. The importance of discriminator

The generative capabilities of GAN have important implications for further improving the quality of reconstruction results. The proposed method DADIGAN is trained by removing the discriminator, recorded as w/o Dis. The quantitative comparisons results are presented in Fig. 11, where our proposed method improves the results of the w/o Dis on four evaluation metrics. Moreover, the visual results shown in Fig. 12 intuitively demonstrate that the proposed method can distinctly improve the performance of the reconstructed images compared with the w/o Dis.

5.2. Analysis of the number of stage T

To test the performance of our proposed network DADIGAN under different numbers of stage T. We set different numbers of stage T to train the DADIGAN. The visual reconstruction results in Fig. 13 show how the number of stages T influences result of cloud removal. Particularly, the SAM error maps between the reconstruction and target images under different stages T intuitively prove that the optimal results are achieved when T is equal to 3, which can also be evidenced by the quantitative evaluation results in Table 2.

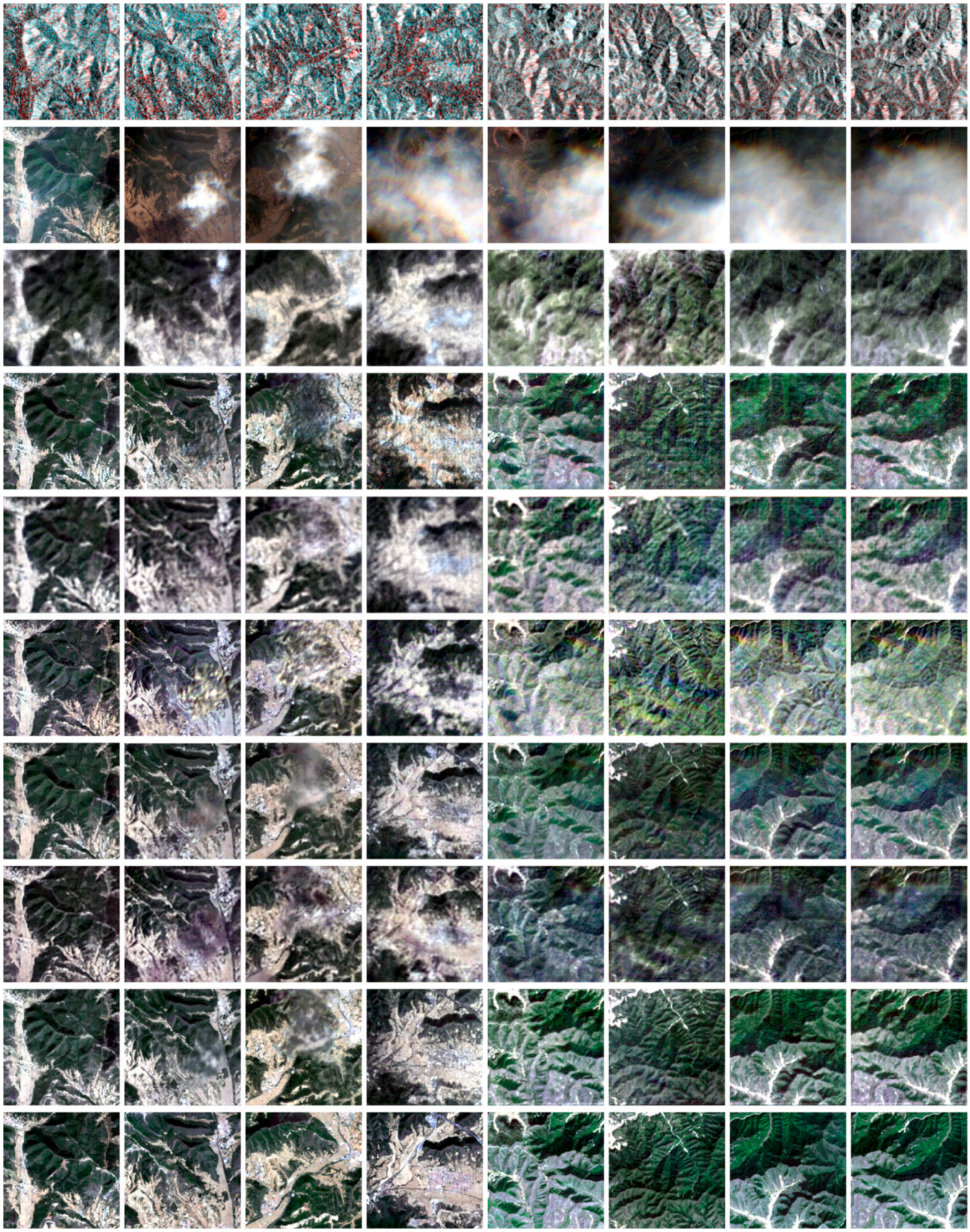


Fig. 10. Cloud and shadow removal results on the SEN12MS-CR dataset under different cloud coverage levels. For each example sample, from the top row to the bottom row represent SAR images, cloud images, the results of Pix2pix, McGAN, SAR-Opt-cGAN, Dsen2-CR, GLF-CR, Align-CR, ours, and the cloud-free images.

5.3. Analysis of the GAN loss weight

We use multiple loss functions to jointly optimize the parameters of our proposed network, which include the adversarial loss, L1loss and KL divergence loss. Two tuning parameters λ_1 and λ_2 are applied to balance the contribution of the L1loss and KL divergence loss. Fig. 14 presents how the model behaves differently when λ_1 and λ_2 take different values. Apparently, the proposed method provides a remarkable improvement in accuracy which $\lambda_2 = \lambda_2 = 100$.

5.4. Analysis of the computational efficiency

Floating point operations per second (Flops) and parameters of the network are critical factors that affect the application of the network in resource-constrained environments. In order to comprehensively evaluate the performance of our proposed method in terms of computational resources and inference time, We calculate the number of Flops, the learnable parameters, and the inference time for all methods on the same device (Nvidia RTX 4090) using data of size 256*256,

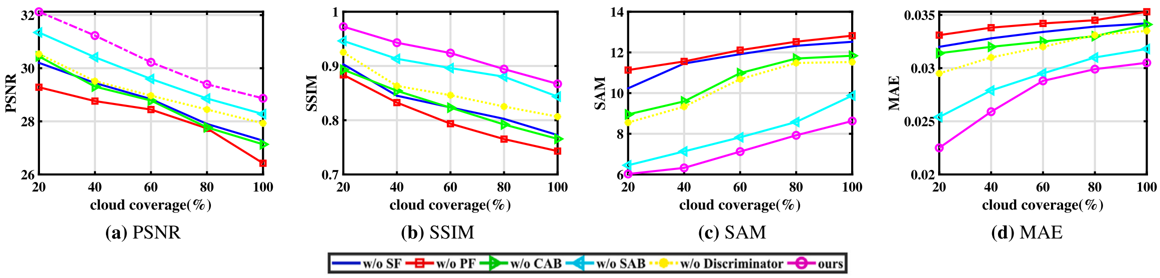


Fig. 11. Quantitative evaluation results of ablation experiments.

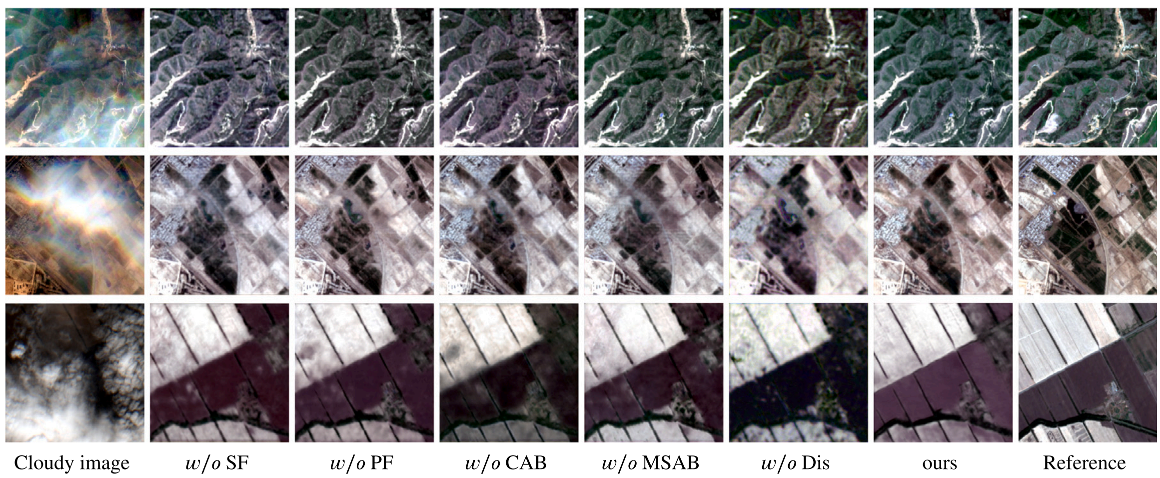


Fig. 12. Cloud and shadow removal results on the SEN12MS-CR dataset under different land cover types.

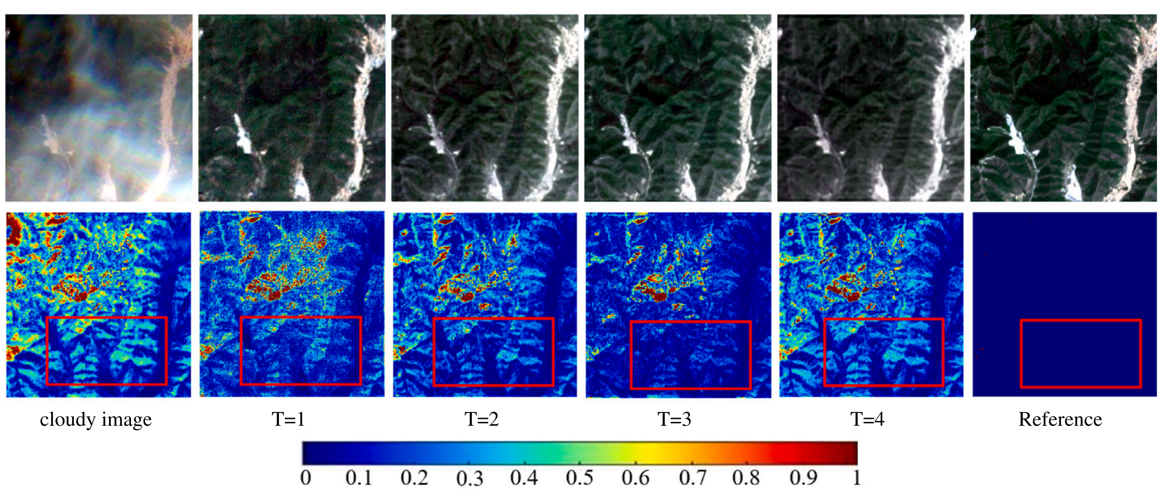


Fig. 13. A visual comparison result of different the Number of Stages T.

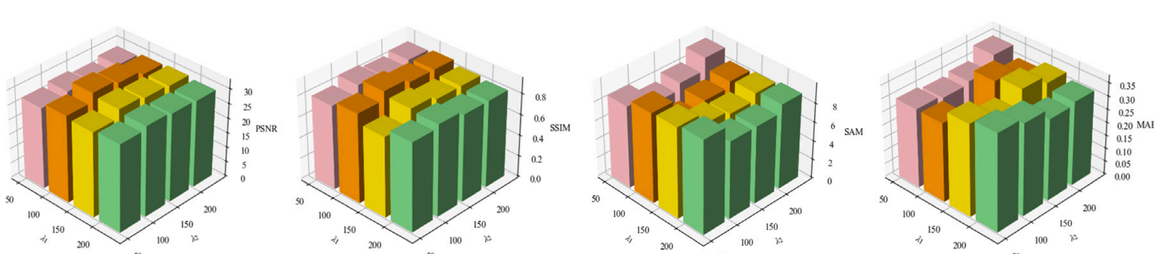


Fig. 14. Relationship between the performance of the proposed method and tuning parameters.

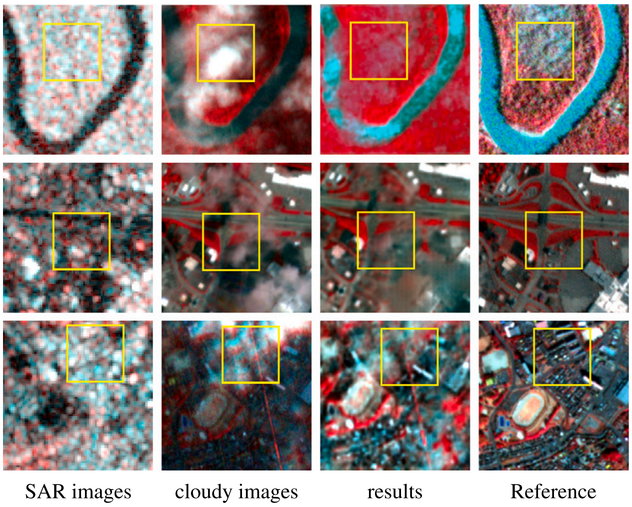


Fig. 15. Three typical cases show the model performance when SAR images struggle. These cases include river valleys, urban areas, and severe speckle noise.

Table 3
The computational efficiency of all methods.

Method	Flops	Params	Time
Pix2pix	**61.0GMac**	11.45M	1.61 ms
McGAN	72.27GMac	54.43M	**1.58 ms**
SAR-opt-cGAN	91.68GMac	218.03M	1.64 ms
Dsen2-CR	1241.68GMac	18.94M	2.33 ms
GLF-CR	244.63GMac	14.83M	240.00 ms
Align-CR	174.92GMac	43.52M	185.00 ms
Ours	191.58GMac	**1.74M**	5.80 ms

respectively. Distinctively, the GAN-based methods only compute the learnable parameters of the generators to ensure fairness. When calculating the model complexity of all methods, 13-channel optical images and 2-channel SAR images are used as network inputs. The inference time is the average of all the test data. In addition, since GLF-CR can only process $128 * 128$ sized data, it takes four times as long as the fusion time for $128 * 128$ sized data. The quantitative results in Table 3 show that our proposed method has the fewest parameters compared to the comparison methods, while the Flops and inference time are in the middle among all methods due to the iterative training strategy adopted by the shared and private feature extraction modules, but within an acceptable range. Furthermore, Align-CR and GLF-CR have longer inference time mainly due to the Transformer-based backbone network.

5.5. The impact of modal heterogeneity on the performance

Considering the modal heterogeneity of optical and SAR images, we design a shared and private feature fusion strategy to reconstruct cloud-free optical images in this paper. Although our proposed method has achieved good performance in most scenarios, SAR images cannot fully capture the texture details of optical images due to the modal heterogeneity in some complex scenarios. When the optical image is covered by clouds, the performance of our proposed method may degrade. In addition, the speckle noise of SAR images in some complex scenes will also affect the performance of the model. We show some examples where SAR images struggle in Fig. 15, which include river valleys, urban areas, and severe speckle noise where the model performance degrades.Specifically, the visual results in the first-row show that when the SAR image cannot fully capture the marshland area on the red mudflat in the optical image, the reconstruction result of the yellow rectangular box in the cloud removal result is different from the reference image. The visual results in the second-row show that the

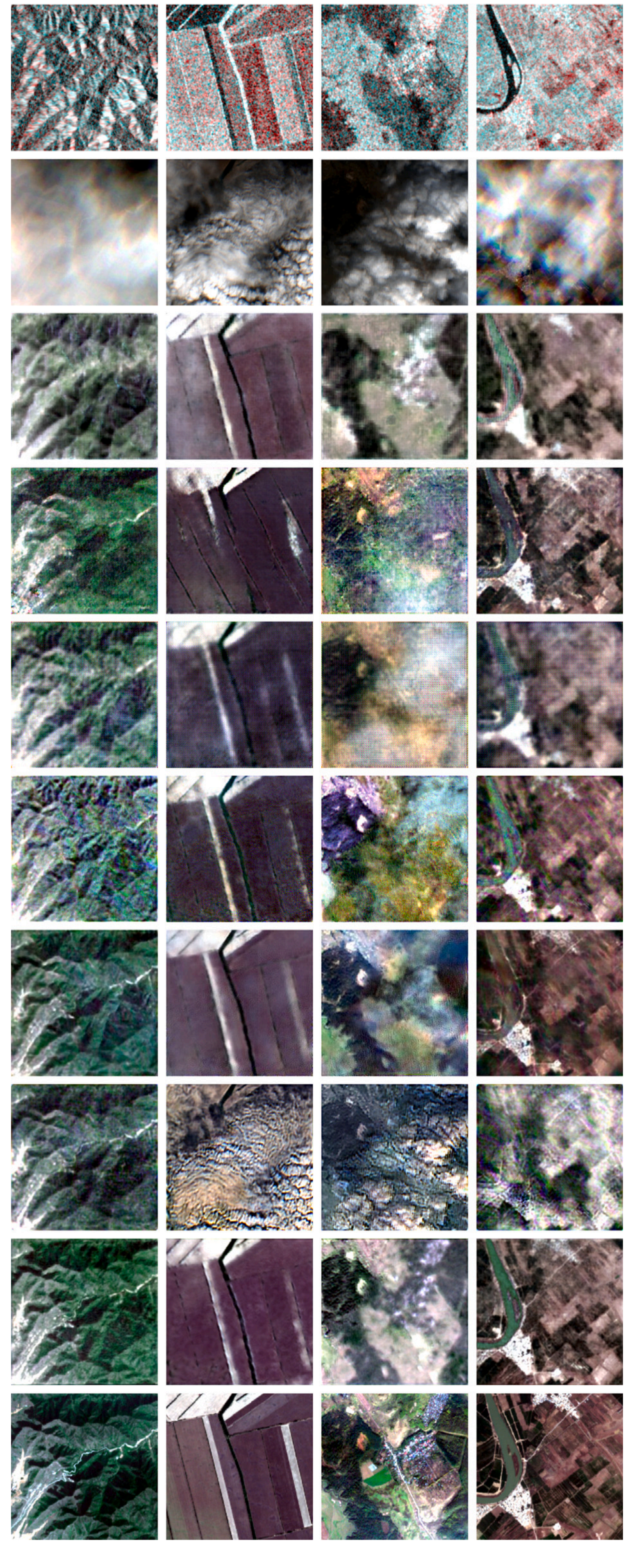


Fig. 16. Example results of our proposed method on the images completely obscured by clouds. For each example sample, from the top row to the bottom row represent SAR images, cloud images, the results of Pix2pix, McGAN, SAR-Opt-cGAN, Dsen2-CR, GLF-CR, Align-CR, ours, and the cloud-free images.

SAR image in a complex urban scene cannot capture the texture details of the cloud-covered area, and the details of reconstruction result of the yellow rectangular box in the cloud removal result are incomplete. The visual results in the third-row show that when the imaging quality of the SAR image in a complex scene is severely affected by coherent speckle noise, the reconstruction result in the yellow rectangular box in the cloud removal result is of poor quality. In the future, we will try to design a comprehensive modal fusion strategy to further improve the model performance.

5.6. The performance of the model when completely occluded by cloud and shadow

Under the circumstance of complete occlusion, single image cloud removal and multi-temporal cloud removal methods become ineffective. The performance of our proposed method and the comparison methods under different land cover types is tested when optical images are completely covered by cloud and shadow. The visual reconstruction results in Fig. 16 show that the spatial information is effectively recovered even under complete cloud and shadow coverage. The reconstructed images show obvious spectral distortion because SAR images lack spectral information. It is obvious that although our proposed method achieves the best performance, there is still much room for improvement in its performance compared to the performance without full cloud coverage, and we will try to improve the performance of optical image reconstruction under full cloud coverage by introducing a SAR colorization [61] stage before fusing optical and SAR images in our future work.

6. Conclusion

In this article, a dual attention blocks-based disentangled iterative GAN for cloud and shadow removal, named as DADIGAN, is proposed. In DADIGAN, we first construct a convolutional sparse coding-based observation model, the PGDA is utilized to optimize the model, which is achieved through a series of iterative steps. These iterative steps are unfolded into neural network modules to explicitly extract shared and private features from cloudy and SAR images. After that, the PDAFM is designed to fuse the shared and private features from different modalities while avoiding adverse reactions due to modal heterogeneity. Finally, the RB decodes the fused features to reconstruct high-fidelity cloud-free images. Experimental results on the SEN12MS-CR dataset have demonstrated that DADIGAN yields the best cloud removal results under different land cover types and various levels of cloud coverage compared with other approaches.

However, the quality of reconstructed image is not satisfactory when the optical image is completely covered by cloud and shadow. Next, we will attempt to use contrast learning to further improve the quality of reconstructed image when the optical image is completely obscured by cloud and shadow.

CRediT authorship contribution statement

Zhongchen He: Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Peng Wang:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yunkun Zou:** Writing – review & editing, Project administration, Methodology, Data curation. **Bo Huang:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Daiyin Zhu:** Writing – review & editing, Visualization, Validation, Supervision. **Harry F. Lee:** Visualization, Validation. **Henry Leung:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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